

Review of Portfolio Optimization Region

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I. INTRODUCTION

Portfolio optimization is a core issue in the field of quantum finance. The main goal is to find an ideal balance between risk and return. The classical theory was proposed by Harry Markowitz, in which the optimal portfolio is expressed as an mean-variance optimization problem[1].The importance of this framework remains because it gives a clear mathematical definition of the balance between risk and return. However, with the development of the financial market, the limitations of these traditional methods are increasingly emerging.

First, financial data itself has a high-dimensional nature. A portfolio may contain a large number of assets, factors, indicators and macroeconomic variables. Second, asset returns are usually non-linear and non-steady. Therefore, models based on historical data fitting may fail when the market changes. Third, financial decision-making is usually gradual. Current portfolio decisions will affect future wealth levels, risk exposure, transaction costs and subsequent rebalancing choices. Fourth, assets are not independent of each other. The correlation, industry relationship, common factors and risk transmission paths of these assets will change over time.

The above limitations explain why recent research has shifted from static optimization to artificial intelligence-based financial decision-making. Deep learning methods can be used to model nonlinear

financial time series and extract complex market characteristics from them[2], [3].However, deep learning models are usually used to predict tasks. This kind of model can more accurately estimate future price trends or returns, but it cannot directly provide guidance for optimal investment decision-making. This framework of separation of forecasting and decision-making leads to a gap between prediction accuracy and actual combination performance. A model may make accurate predictions of prices, but still cannot formulate a profitable or risk-controllable asset allocation strategy.

Compared with deep learning (DL), reinforcement learning makes up for this shortcoming more directly. RL is not limited to predicting future returns, but learns a strategy to map the market state into portfolio behavior. The goal pursued is not only the accuracy of the forecast, but also to achieve the cumulative investment performance under market constraints. Therefore, this literature review focuses on reinforcement learning, with particular emphasis on deep reinforcement learning as its contemporary extension for financial decision-making and quantum finance applications.

II. DEEP REINFORCEMENT LEARNING: WHAT IT IS

Reinforcement learning is a branch of machine learning regarding sequential decision-making. In an RL problem, an agent observes the current state of an environment and then chooses an action. After the

action, it receives a reward and moves to a new state. Through repeated interaction with environment, the agent learns a policy that maximizes expected cumulative reward [4].

In portfolio optimization, the environment is financial market. An agent is the portfolio manager or trading model. The state may include asset prices, returns, technical indicators, volatility, market factors, macro variables, or quantum-finance indicators. The action usually represents portfolio weights, trading signals, position sizes, or buy-sell-hold decisions. The reward measures quality of the action, such as portfolio return, risk-adjusted return, Sharpe ratio, drawdown control, transaction-cost-adjusted profit, or a multi-objective combination of these criteria.

The key advantage of RL is that it treats investment management as a dynamic control problem. Portfolio optimization is not a one-time prediction task. It requires continuous rebalancing under uncertainty of the market. RL is therefore suitable for financial engineering problems as decisions, market feedback, and risk exposure evolve over time.

III. HOW DEEP REINFORCEMENT LEARNING WORKS IN PORTFOLIO OPTIMIZATION

A typical RL-based portfolio optimization framework usually contains four components: state representation, action design, reward function, and policy learning.

A. State Representation

The state describes those information available to agent before making a decision. Basic state variables include historical prices, returns, moving averages, volatility, and technical indicators. More advanced models may also include financial factors, asset correlations, macroeconomic signals, or graph-based representations of asset relationships. A strong state representation is important because agents can only learn from the information encoded in the state.

B. Action Space

The action defines what the agent can do. In discrete trading tasks, actions may be buy, sell, or hold. In portfolio allocation tasks, actions are usually continuous portfolio weights. These continuous actions are more realistic because a portfolio manager must decide not only whether to invest in an asset but also how much capital to smartly allocate. Deep deterministic policy gradient (DDPG) [5], twin delayed DDPG (TD3) [6], proximal policy optimization (PPO) [7], and related autoregressive policy optimization methods [8], [9], [7] are frequently used because they can handle high-dimensional or continuous decision spaces.

C. Reward Function

The reward function defines investment goal. A simple reward may be the portfolio return. However, return-only rewards often lead to unstable or extremely risky policies. Recent studies therefore introduce risk-adjusted and multi-objective rewards. These rewards may include controllable risk terms in multi-objective formulations [10], beta-exposure or factor-based risk terms [11], and volatility penalties, maximum drawdown, transaction costs, turnover constraints, or other risk-adjusted objectives [12]. Reward design is one of the most important issues in financial RL because it determines whether the learned policy is merely profitable in backtests or also stable and risk-aware.

D. Policy Learning

The agent learns a policy by interacting with historical or simulated market data. During training, the model updates its parameters to improve expected cumulative reward. In portfolio optimization, this process is difficult because financial data are noisy, market patterns change, and the number of truly independent observations is limited. As a result, RL models may suffer from slow learning and high sample complexity [4], unstable convergence [13], and overfitting during backtest or weak out-of-sample robustness

[12], [13]. These challenges motivate recent research on constrained policy optimization [9], autoregressive allocation [8], [9], graph-based policy learning [14], [15], and multi-agent architectures [15], [16].

IV. LATEST RESEARCH AND CONTEMPORARY TECHNOLOGY

Instead of simply maximizing return, the main trend of recent RL research in portfolio optimization is to make RL models more realistic and sensitive to risks. This is achieved specifically by adding some constraints directly into the learning process.

A. *Autoregressive and Constrained RL*

Autoregressive DRL models consider portfolio weights as a sequence of dependent decisions rather than independent output variables. This structure allows policy network to model relationships among assets and allocation constraints more effectively [8]. By designing policies that explicitly satisfy budget or linear requirements, these autoregressive approaches address the practical demands of professional trading [9]. Such frameworks are important because real portfolios must obey strict constraints such as full investment and transaction cost limits.

B. *Risk-Adjusted and Multi-Objective RL*

To avoid only optimizing raw return, more recent studies incorporate multiple financial objectives into the reward function. The MILLION framework formulates portfolio management as a general multi-objective problem with controllable risk [10]. Multi-reward RL methods also combine return, volatility, drawdown, and other indicators to improve the balance between profitability and stability [12]. Factor-based DRL introduces financial factor information into the reward or state design, which helps the policy respond to long-term market exposure rather than only short-term price movements [11].

C. *Graph-Based and Multi-Agent RL*

The core idea of graph-based methods is to represent assets as nodes in graphs, and correlations,

sector relationships, common factors or information transmission relationships as edges. Through modeling dynamic relationships among assets with graph attention and a heterogeneous multi-agent framework, graph attention-based heterogeneous multi-agent DRL ensure that the policy can adaptively adjust the portfolio according to both local asset information and global market structure [15]. Compared with ordinary neural network, GNN can explicitly utilize asset network structure and avoid simply concatenating all asset features before feeding them into a single policy network. This design improves model's ability to represent cross-asset dependencies and the ways risk spreads through a portfolio.

By shifting to multi-agent architectures, optimization task is further decomposed to separate entities that either cooperate or compete. Rather than a single model handling everything, individual agents focus on specific domains, such as a particular asset class, sector, or trading function. Through centralized training with decentralized execution (CTDE), collective investment strategy emerges from agents' interactions. This approach is effective for multi-asset markets where information is unevenly distributed and risk factors are highly varied. As a trade-off, multi-agent methods model local asset behavior in more detail, but they also introduce additional complexity in policy coordination and training stability.

D. *Inverse RL and Heuristic-Guided Reward Learning*

Another promising direction is integrating inverse reinforcement learning (IRL) with multi-objective rewards and graph-based policy learning. IRL extracts reward signals directly from expert behavior or historical decision patterns, rather than relying on man-designed objectives. This heuristic-guided approach reduces subjectivity within reward function engineering, which is a persistent challenge in financial RL. However, reward learning alone does not fully address the relations in portfolio optimization, because the value of one asset is often affected by

other assets. By integrating graph structure into policy learning, the model can better capture dependency and diversification effects between different assets [14].

E. Remaining Limitations

Contemporary RL methods have improved portfolio optimization, but several limitations still remain. A primary concern is that RL models are sensitive to reward design and market simulation assumptions. Many models show strong backtest performance but weaker robustness when encountering new data. This suggests that the learned policies just capture historical patterns or noise rather than stable market patterns. Additionally, factors like transaction costs, liquidity, and market impact are still simplified in many studies. Thus, the reported returns may be difficult to realize in actual trading environments. Finally, complex architectures improve representation power but are hard to reproduce. For instance, deep neural networks and multi-agent systems can capture richer market structures, but their decision logic is difficult to explain and their results depend heavily on implementation details.

V. INTEGRATION WITH QUANTUM FINANCE AND FINANCIAL ENGINEERING APPLICATIONS

In this section, we concern how reinforcement learning can be integrated with Quantum Finance and applied to financial engineering. Basically, this integration can be understood through Quantum Price Levels, quantum-finance-based risk signals, and multi-agent decision systems.

A. State-Level Integration

Quantum Price Levels (QPL) are price-level indicators derived from Quantum Finance theory. The basic idea is to describe financial price movements by analogy with particles moving between energy levels. In this view, an asset price may stay near a relatively stable level when market conditions are calm, but may jump to a higher or lower level when market forces

change. These levels can be seemed as important price zones in financial markets.

At the state-representation level, QPL can be used as an external input to an RL agent. A standard RL model usually observes historical returns, prices, volatility, and technical indicators. After integrating QPL, the agent can also observe whether the current price is close to a high-risk price level, a support-like zone, or a resistance-like zone. This enriches the market state observed by the agent. Instead of learning only from raw historical data, the agent learns additional information about the structural position of the asset price.

This integration is useful because financial markets are noisy and not stationary. Purely data-driven RL models may struggle to distinguish temporary fluctuations from meaningful changes of market patterns. QPL-based state variables can provide an interpretable market-structure signal and may help the agent make more stable allocation or trading decisions.

B. Objective-Level Integration

QPL can also be integrated into the reward function or constraints for risk-control. A return-only reward may encourage model to take excessive risks, because the agent is rewarded mainly for short-term profit. If QPL indicates that the asset price is entering a risky region, the reward function can penalize aggressive position increases or encourage position reduction.

This distinction is important because when QPL is used as a state variable, it tells the agent what the market structure looks like. When QPL is used in reward shaping, it tells the agent which behaviors should be encouraged or discouraged under that market structure. For example, an RL agent may receive a lower reward if it increases exposure near a risky QPL zone, even if the short-term return is positive. This design can make the learned policy more risk-aware.

The augmented DDPG approach integrates QPL-based trading strategies into dynamic portfolio optimization[5]. This type of design successfully addresses the limitation of conventional RL that risk control is often learned only indirectly from historical rewards. By introducing QPL signals into the reward or constraint structure, agent can receive more explicit guidance before large losses occur.

C. Architecture-Level Integration

A third form of integration appears at the system-architecture level. Quantum Finance can be combined with fuzzy reinforcement learning and multi-agent trading systems. QF-FRL combines Quantum Finance, fuzzy reasoning, and a multi-agent trading framework [16]. In this type of system, different agents or modules can perform different functions. For instance, one agent may focus on trading decisions that seeking for returns, while another agent may monitor risk exposure based on information of Quantum Finance and fuzzy rules.

Fuzzy logic is useful in this context because many financial signals do not have clear binary boundaries. Market risk, trend intensity and trading confidence usually change gradually, rather than simple fluctuations. With the help of fuzzy reasoning, these vague market conditions can be transformed into more flexible risk control rules. When fuzzy reasoning is combined with reinforcement learning, the system can learn from market feedback while still retaining rule-based interpretability.

The multi-intelligent body structure further separates alpha revenue generation from risk management. This arrangement is closer to the practical operation in financial engineering, because in financial engineering, an investment system usually includes both transaction modules and independent risk control modules. Although this multi-intelligent architecture can make the decision-making process more modular, it comes at the cost of bringing challenges such as coordination, communication and computational complexity

between intelligent bodies.

D. Summary of applications in Financial Engineering

The integration of RL and Quantum Finance has multiple potential applications.

First, it can support dynamic portfolio allocation. RL determines portfolio weights over time, while QPL or other quantum-finance indicators provide structural price-level information. Second, it can improve algorithmic trading. RL can learn trading policies, and QPL can guide entry, exit, and position sizing. Third, it can strengthen risk management. Quantum-finance signals and fuzzy rules can be used to limit exposure when prices approach high-risk regions. Fourth, it can support hedging and derivative-related decisions. If QPL signals a change in price-state structure, the RL agent can adjust hedge ratios or rebalance option-related positions. Fifth, it can improve interpretable decision support. Portfolio managers may better understand an RL recommendation if it is linked to explicit price levels and risk-control rules.

Overall, Quantum Finance provides external structural information, while RL provides adaptive decision-making. Their combination is promising because it connects the state of market with dynamic policy optimization.

VI. CONCLUSION

This paper discussed reinforcement learning as an AI technology for portfolio optimization and Quantum Finance. Unlike traditional deep learning methods that mainly predict returns or prices, RL directly learns investment decisions through interaction with a market environment. The sequential decision making process and prioritization of cumulative rewards over one-step accuracy make RL an ideal method for portfolio management.

Recent research has improved RL portfolio models through autoregressive policies, constraint-aware allocation, multi-objective rewards, factor-based reward design, graph-based representation, multi-agent

systems, and inverse reinforcement learning. These developments show that the field is moving toward more realistic and risk-aware financial decision systems. However, RL still faces challenges in sample efficiency, convergence stability, reward design, interpretability, and robustness under changing market patterns.

The integration of RL with Quantum Finance provides a useful direction for financial engineering. Quantum Price Levels and fuzzy risk-control modules can provide external market-structure signals for RL agents. Future research should focus on building unified frameworks that combine adaptive policy learning, explicit risk control, market-structure representation, and robust out-of-sample validation.

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